

Andrew Zhao

azhao@sandia.gov | GitHub.io | Google Scholar

EDUCATION	University of New Mexico, Albuquerque, NM, USA <i>Department of Physics and Astronomy</i> <i>Center for Quantum Information and Control</i> PhD, Physics (with distinction) Advisor: Akimasa Miyake Dissertation: <i>Learning, Optimizing, and Simulating Fermions with Quantum Computers</i> Chairman's Award for Best Dissertation, 2024 MSc, Physics	Aug 2018 to May 2024 Defended Nov 2023 Jul 2021
	University of Maryland, College Park, MD, USA <i>Department of Physics</i> <i>University of Maryland Honors College (Gemstone Program)</i> BSc, Physics	Sep 2013 to May 2018
EXPERIENCE	Postdoctoral Research Fellow, Sandia National Laboratories (Livermore, CA) <i>Quantum Algorithms and Applications Collaboratory</i> <i>Gil Herrera Fellowship in Quantum Information Science</i> Mentors: Andrew Baczewski and Ojas Parekh Student Researcher, Google (San Francisco, CA) <i>Google Quantum AI</i> Host: Nicholas Rubin Graduate Research Assistant, University of New Mexico (Albuquerque, NM) <i>Department of Physics and Astronomy</i> <i>Center for Quantum Information and Control</i> Advisor: Akimasa Miyake Graduate Teaching Assistant, University of New Mexico (Albuquerque, NM) <i>Department of Physics and Astronomy</i> Undergraduate Research Assistant, University of Maryland (College Park, MD) <i>Joint Center for Quantum Information and Computer Science</i> Mentor: Shelby Kimmel	Feb 2024 to present May 2022 to Dec 2022 Jan 2019 to Dec 2023 Aug 2018 to May 2019 Feb 2016 to May 2017
GRANTS	"Investigating quantum algorithms and complexity for applications to physical simulation" US\$265,000 Project PI. Funded by the Laboratory Directed R&D program at Sandia National Laboratories, Computing & Information Sciences Division, as part of the Gil Herrera Fellowship. Supported two graduate student interns.	Awarded 2024
PUBLICATIONS	Peer-reviewed articles A. Zhao, "Learning the Structure of Any Hamiltonian from Minimal Assumptions," <i>Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)</i>, 1201, 2025. A. Zhao, N. C. Rubin, "Expanding the reach of quantum optimization with fermionic embeddings," <i>Quantum</i> 8, 1451, 2024. A. Zhao, A. Miyake, "Group-theoretic error mitigation enabled by classical shadows and symmetries," <i>npj Quantum Information</i> 10, 57, 2024. R. Babbush, W. J. Huggins, D. W. Berry, S. F. Ung, A. Zhao, D. R. Reichman, H. Neven, A. D. Baczewski, J. Lee, "Quantum simulation of exact electron dynamics can be more efficient than classical mean-field methods," <i>Nature Communications</i> 14, 4058, 2023.	

- A. K. Daniel, Y. Zhu, C. H. Alderete, V. Buchemmavari, A. M. Green, N. H. Nguyen, T. G. Thurtell, **A. Zhao**, N. M. Linke, A. Miyake, “Quantum computational advantage attested by nonlocal games with the cyclic cluster state,” *Physical Review Research* **4**, 033068, 2022.
- **A. Zhao**, N. C. Rubin, A. Miyake, “Fermionic Partial Tomography via Classical Shadows,” *Physical Review Letters* **127**, 110504, 2021.
- **A. Zhao**, A. Tranter, W. M. Kirby, S. F. Ung, A. Miyake, P. J. Love, “Measurement reduction in variational quantum algorithms,” *Physical Review A* **101**, 062322, 2020.

Pre-prints

- A. Christensen, **A. Zhao**, “Learning fermionic linear optics with Heisenberg scaling and physical operations,” [quant-ph/arXiv:2602.05058](#), 2026.
- B. Harrison, V. Iyer, O. Parekh, K. Thompson, **A. Zhao**, “Fermionic Insights into Measurement-Based Quantum Computation: Circle Graph States Are Not Universal Resources,” [quant-ph/arXiv:2510.05557](#), 2025.

Patents

- N. C. Rubin, **A. Zhao**, “Solving quadratic optimization problems over orthogonal groups using a quantum computer,” [US Patent US 2024 0296367 A1](#), 2024.

TALKS

Invited

- “Hamiltonian learning: A survey of recent progress” Mar 2025
University of Sherbrooke Institut quantique Colloquium, *Sherbrooke, QC*
- “Group-theoretic error mitigation enabled by classical shadows and symmetries” Nov 2023
University of Oxford Quantum Technology Theory Seminar, *Virtual*
- “Quantum relaxation for quadratic programs over orthogonal matrices” Oct 2023
INFORMS Annual Meeting, *Phoenix, AZ*
- “Efficient and robust learning of quantum many-body properties with classical shadows” Jun 2023
Sandia Quantum Computer Science Seminar, *Virtual*
- “Efficient and robust learning of fermionic reduced density matrices with classical shadows” Jan 2023
UC Berkeley Quantum Many-body Seminar, *Virtual*

Contributed

- “Learning the structure of any Hamiltonian from minimal assumptions” Jun 2025
ACM STOC, *Prague, CZ*
- “Learning the structure of any Hamiltonian from minimal assumptions” Mar 2025
APS Global Physics Summit, *Anaheim, CA*
- “Learning the structure of any Hamiltonian from minimal assumptions” Feb 2025
QIP Conference, *Raleigh, NC*
- “Nearly Heisenberg-limited Hamiltonian learning with few assumptions” Oct 2024
SQuInT Workshop, *Broomfield, CO*
- “Group-theoretic error mitigation enabled by classical shadows and symmetries” Oct 2023
SQuInT Workshop, *Albuquerque, NM*
- “Calibration-free quantum error mitigation with classical shadows” Mar 2023
APS March Meeting, *Las Vegas, NV*
- “Fermionic partial tomography via classical shadows” Mar 2021
APS March Meeting, *Virtual*
- “Measurement reduction in variational quantum algorithms” Mar 2020
APS March Meeting, *Denver, CO → Virtual*

TEACHING

Courses

- General Physics (PHYC 151), Graduate Supplemental Instructor Spring 2019
University of New Mexico, Department of Physics and Astronomy
- General Physics Laboratory (PHYC 151L), Graduate Lab Instructor Fall 2018
University of New Mexico, Department of Physics and Astronomy

Guest Lectures

- “Classical shadows are both simpler and more complicated than you think” Mar 2025
University of Sherbrooke Institut quantique seminar, *Sherbrooke, QC*

- “Quantum metrology and distributed quantum sensing” Jun 2023
UNM CQuIC Summer Course, *Albuquerque, NM*
- “Random unitary evolution, t -designs, and applications to quantum chaos” Jun 2021
UNM CQuIC Summer Course, *Virtual*
- “Continuous symmetries and $O(N)$ models” Jun 2020
UNM CQuIC Summer Course, *Virtual*
- “Simple Lie groups and simple Lie algebras” Jun 2019
UNM CQuIC Summer Course, *Albuquerque, NM*

ADVISING

Sandia Graduate Interns

- Ramya Bhaskar, *University of Washington, PhD student* Sep 2024 to Dec 2025
– Now a postdoc at Sandia
- Aria Christensen, *Ohio State University, PhD student* Aug 2024 to present

PROFESSIONAL SERVICE

Journal Referee

- Physical Review Letters • Physical Review A • Physical Review Research • PRX Quantum • Journal of Chemical Theory and Computation • Nature Communications • npj Quantum Information • Quantum • Communications in Mathematical Physics

Conference Referee

- QIP • TQC • QSim • QCTiP

Grant Proposal Reviewer

- Sandia LDRD program